

Learning Objectives

This chapter contains the following topics:

- a training set containing patterns with missing attribute values
- developing the mathematical model for the ratio of information gain provided by an attribute that has missing values
- building a decision tree when some of the training patterns have missing attribute values
- associating each non-leaf node with an attribute being examined, and each leaf node with the probability of a pattern belonging to a particular class or being rejected
- classifying a given pattern using a decision tree built from training patterns with missing attribute values
- predicting the error rate on recall sets based on the error rate observed on the training set
- pruning a decision tree based on the predicted error rate
- windowing of training sets in an attempt to reduce the computation required in building a decision tree

3

Decision Trees: Extensions

If attribute A has $n > 1$ possible values, it will split V_0 into sets $V_1, V_2, \dots, V_n$. Those patterns in V_0 whose value of attribute A is missing will not be in any of the sets V_i . For $1 \leq k \leq n$ and $0 \leq i \leq n$, let $Z(k, i)$ be the number of training patterns of class C_k in the set V_i . Let $Z(i)$ be the number of patterns in V_i . Moreover, let

$$Z'(0) = Z(0) - Z(n+1)$$

Out of $Z(0)$ patterns in set $V(0)$, we thus have $Z'(0)$ patterns for which the value of attribute A is known. Proceeding as we did for Eqn (2.3) is

3.1 Missing Attribute Values

At times, a few attribute values are missing from some of the patterns in a training set. Say, in an effort to improve the treatments for different kinds of cancer, training set data are collected from many hospitals about their cancer patients: the attributes for each patient may describe personal information about the patient, the symptoms the patient had, the nature of cancer, the treatment received, and the effects of the treatment. Suppose, owing to oversights in recording the data at the hospital, some attribute values are missing. For example, for one patient, the age—which may be considered to decide the kind of treatment to be prescribed—might not have been recorded; for another patient, the size of the cancerous tumour might not have been recorded.

We can delete from the training set any patterns with missing attribute values, but if there are many such patterns, the training set could become small, reducing the generality of the decision tree built. Often the patterns with missing attribute values are allowed to remain in the training set. The equations to evaluate the attribute's ratio of information gain, described in Section 2.5, are then modified as follows.

Let $C_1, C_2, \dots, C_m$ be the $m \geq 1$ classes to which the training patterns belong. Suppose V_0 , a subset of the training set V , is associated with a node y_0 of a decision tree we are building. We want to evaluate the ratio of information gain for attribute A at node y_0 , but the value of attribute A is missing for some of the patterns in V_0 .

If attribute A has $n > 1$ possible values, it will split V_0 into sets $V_1, V_2, \dots, V_n$. Those patterns in V_0 whose value of attribute A is missing will not be in any of the sets V_1 to V_n . We will temporarily put such patterns in a set V_{n+1} . For $1 \leq k \leq m$ and $0 \leq i \leq n$, let $Y(k, i)$ be the number of training patterns of class C_k in the set V_i . For $0 \leq i \leq (n + 1)$, let $Z(i)$ be the number of patterns in V_i . Moreover, let

$$Z'(0) = Z(0) - Z(n + 1)$$

Out of $Z(0)$ patterns in set $V(0)$, we thus have $Z'(0)$ patterns for which the value of attribute A is known. Proceeding as we did for Eqn (2.1) in

Section 2.5, we obtain the entropy of the set V_0 as (remember all logs are to the base 2)

$$I(V_0) = - \sum_{k=1}^m \frac{Y(k, 0)}{Z'(0)} \log \frac{Y(k, 0)}{Z'(0)} \quad (3.1)$$

Similarly, as we did for Eqn (2.2) in Section 2.5, we obtain the entropy of attribute A for set V_0 as

$$I_A(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z'(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} \quad (3.2)$$

The patterns that are put in V_{n+1} (that is, those whose value of attribute A is missing) are not used in Eqns (3.1) and (3.2), because A does not provide any information about their class. Nonetheless, to obtain the gain in information caused by attribute A , we need to multiply the difference between $I(V_0)$ and $I_A(V_0)$ by $Z'(0)/Z(0)$, the probability of the value of attribute A being known in the set V_0 . Accordingly, Eqn (2.3) of Section 2.5 is altered to the following:

$$g_A(V_0) = \frac{Z'(0)}{Z(0)} [I(V_0) - I_A(V_0)] \quad (3.3)$$

Attribute A has effectively split V_0 into $n+1$ sets, V_1 to V_{n+1} . Then, by adapting Eqn (2.4) of Section 2.5, the split information of attribute A becomes

$$S_A(V_0) = - \sum_{j=1}^{n+1} \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} \quad (3.4)$$

When $Z'(0) = Z(0)$ (that is, there are no patterns with missing values of attribute A), Eqns (3.1) to (3.4) become identical to Eqns (2.1) to (2.4). In other words, Eqns (2.1) to (2.4) are special cases of Eqns (3.1) to (3.4), respectively. The *ratio of information gain* of attribute A for set V_0 is then, as in Eqn (2.5) of Section 2.5,

$$G_A(V_0) = \frac{g_A(V_0)}{S_A(V_0)} \quad (3.5)$$

We want to maximize this ratio of information gain. For a given non-leaf node in the decision tree, we calculate this ratio for each attribute in the

NAME of training pattern	Attributes			Class
	HABIT	EATS	FOOTWEAR	
T1	Gabby	Baked	Clogs	Student
T2	Gabby	Roasted	Sandals	Professor
T3	Gabby	Baked	Sandals	Student
T4	Quiet	Fried	Sandals	Professor
T5	Gabby	Fried	—	Student
T6	Quiet	Baked	Sandals	Student
T7	Gabby	Fried	Sandals	Professor
T8	Quiet	Fried	Clogs	Student

FIGURE 3.1 Training set for recognizing professors and students with the value of FOOTWEAR missing for pattern T5. This training set is adapted from the training set given in Fig. 1.1.

node's candidate set of attributes. While evaluating the ratio of information gain for each attribute in the candidate set, it may happen that some attributes have missing values. The attribute that has the highest ratio is then selected for examination at the node. This criterion of selection can be used in step 6 of the procedure to build decision trees, described in Section 2.2.

As an example, let us build the decision tree for the professor-student training set given in Fig. 3.1. On comparing the training sets given in Figs 1.1 and 3.1, one will notice that the two training sets are the same except that the former has no missing attribute values, whereas the latter has the value of attribute FOOTWEAR missing for pattern T5.

The candidate set of attributes that can be examined at the root consists of HABIT, EATS, and FOOTWEAR. The ratio of information gain for HABIT and EATS can be obtained from Figs 2.13 and 2.14, because their values are the same in the training sets given in Figs 1.1 and 3.1. Accordingly, $G_{\text{HABIT}}(V_0) = 0.0047$ and $G_{\text{EATS}}(V_0) = 0.3233$. Since there exists a pattern with a missing value of FOOTWEAR, we need to use Eqns (3.1) to (3.5), to evaluate $G_{\text{FOOTWEAR}}(V_0)$. Figure 3.2 shows how FOOTWEAR splits those training patterns whose values of FOOTWEAR are known. The ratio

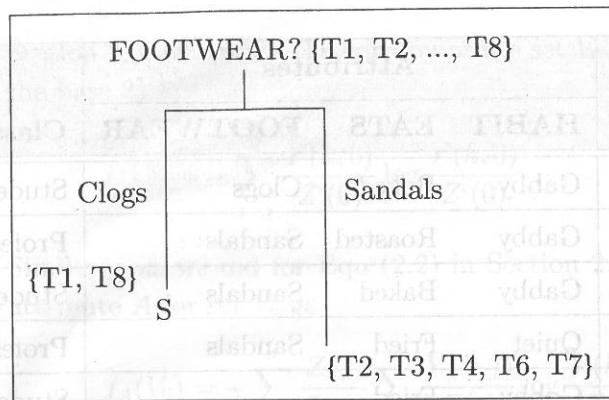


FIGURE 3.2 Examining FOOTWEAR at the root of a decision tree to be built for the professor-student training set given in Fig. 3.1, in which the value of FOOTWEAR is missing for pattern T5. Hence, T5 is not associated with either of the children of the root. Both the patterns associated with the left child of the root belong to the class student, and hence the node becomes a leaf node with label S. Of the patterns associated with the right child of the root, patterns T2, T4, and T7 belong to the class professor; and patterns T3 and T6 belong to the class student.

of information gain for FOOTWEAR, $G_{\text{FOOTWEAR}}(V_0) = 0.1963$, has been calculated in Fig. 3.3. Among $G_{\text{HABIT}}(V_0)$, $G_{\text{EATS}}(V_0)$ and $G_{\text{FOOTWEAR}}(V_0)$, we notice that $G_{\text{EATS}}(V_0)$ is the largest. Hence, we select the attribute EATS to be examined at the root. At the root, the decision tree will, therefore, look like the tree in Fig. 3.4.

From the explanation in the caption of Fig. 3.4, we see that the left and the right children of the root are leaf nodes. The middle child of the root is a non-leaf node, and its candidate set of attributes consists of HABIT and FOOTWEAR. From Figs 3.5 and 3.6 we see that, at the middle child of the root, $G_{\text{FOOTWEAR}}(V_0) = 0.6351$ and $G_{\text{HABIT}}(V_0) = 0$. Since $G_{\text{FOOTWEAR}}(V_0)$ is the larger of the two ratios of information gain, we select FOOTWEAR to be examined at the middle child of the root of Fig. 3.4. The nodes generated by expanding the root's middle child are all leaf nodes, and we obtain a decision tree with a structure similar to the tree in Fig. 2.7, except that pattern T5 will not be associated with the left child of the node at which FOOTWEAR is examined, since the value of FOOTWEAR for pattern T5 is missing. Nonetheless, we will soon see a fraction of pattern T5 being associated with the left child.

To evaluate $G_{\text{FOOTWEAR}}(V_0)$ at the root of a decision tree for the professor-student training set given in Fig. 3.1, in which the value of FOOTWEAR is missing for pattern T5, the following procedure is used. The root is named y_0 . The classes are C_1 (professor) and C_2 (student). Therefore, $m = 2$. Since FOOTWEAR has two possible values (clogs and sandals), root y_0 has two children y_1 and y_2 . Therefore, $n = 2$. To understand the following material easily, you may want to look at Fig. 3.2, since it examines FOOTWEAR at the root.

$V_0 = \{T1, T2, \dots, T8\}$ at node y_0

$Y(1,0) = 3$ (no. of known FOOTWEAR-value patterns in V_0 of class C_1)

$Y(2,0) = 4$ (no. of known FOOTWEAR-value patterns in V_0 of class C_2)

$Z(0) = 8$ (no. of patterns in V_0)

$Z'(0) = 7$ (no. of known FOOTWEAR-value patterns in V_0)

$V_1 = \{T1, T8\}$ at node y_1 , where FOOTWEAR = clogs

$Y(1,1) = 0$ (no. of patterns in V_1 of class C_1)

$Y(2,1) = 2$ (no. of patterns in V_1 of class C_2)

$Z(1) = 2$ (no. of patterns in V_1)

$V_2 = \{T2, T3, T4, T6, T7\}$ at node y_2 , where FOOTWEAR = sandals

$Y(1,2) = 3$ (no. of patterns in V_2 of class C_1)

$Y(2,2) = 2$ (no. of patterns in V_2 of class C_2)

$Z(2) = 5$ (no. of patterns in V_2)

$V_3 = \{T5\}$ (patterns with missing value of FOOTWEAR)

$Z(3) = 1$ (no. of patterns with missing value of FOOTWEAR)

$$I(V_0) = - \sum_{k=1}^m \frac{Y(k,0)}{Z'(0)} \log \frac{Y(k,0)}{Z'(0)} = 0.9852 \quad [\text{from Eqn (3.1)}]$$

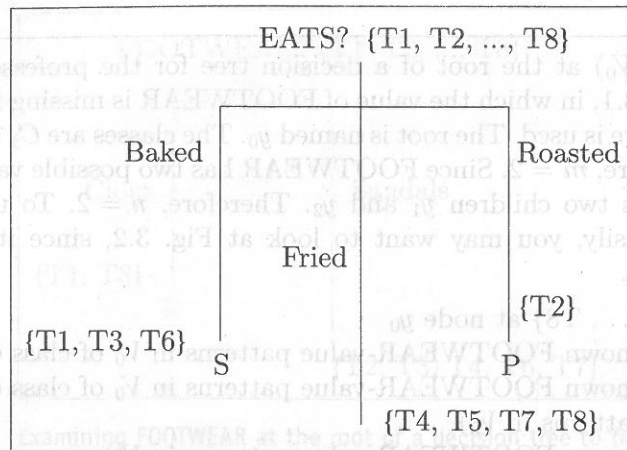
$$I_{\text{FOOTWEAR}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z'(0)} \sum_{k=1}^m \frac{Y(k,j)}{Z(j)} \log \frac{Y(k,j)}{Z(j)} = 0.6936 \quad [\text{from Eqn (3.2)}]$$

$$g_{\text{FOOTWEAR}}(V_0) = \frac{Z'(0)}{Z(0)} [I(V_0) - I_{\text{FOOTWEAR}}(V_0)] = 0.25515 \quad [\text{from Eqn (3.3)}]$$

$$S_{\text{FOOTWEAR}}(V_0) = - \sum_{j=1}^{n+1} \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 1.3 \quad [\text{from Eqn (3.4)}]$$

$$G_{\text{FOOTWEAR}}(V_0) = \frac{g_{\text{FOOTWEAR}}(V_0)}{S_{\text{FOOTWEAR}}(V_0)} = 0.1963 \quad [\text{from Eqn (3.5)}]$$

FIGURE 3.3 Evaluating $G_{\text{FOOTWEAR}}(V_0)$, the ratio of information gain for FOOTWEAR, at the root of a decision tree being built from the professor-student training set given in Fig. 3.1, in which the value of FOOTWEAR is missing for pattern T5, and hence Eqns (3.1) to (3.5) are used above. Set V_0 comprises the eight training patterns. The value of $G_{\text{FOOTWEAR}}(V_0)$ above is less than the value obtained in Fig. 2.15, where no value of FOOTWEAR was missing.

**FIGURE 3.4**

By maximizing the ratio of information gain, EATS has been selected for examination at the root of a decision tree being built for the professor-student training set in which the value of FOOTWEAR is missing for pattern T5. The three patterns associated with the left child of the root belong to the class student, and hence the node becomes a leaf node with label S. Of the patterns associated with the middle child of the root, patterns T4 and T7 belong to the class professor, and patterns T5 and T8 belong to the class student. To build this tree further, we would need to examine HABIT or FOOTWEAR at the middle child of the root. The single pattern associated with the right child of the root belongs to the class professor, and hence the node becomes a leaf node with label P.

Missing attribute values can cause a fraction of a pattern to appear in a set. Let the weight of a pattern in a set be equal to fraction of the pattern in the set. If the full pattern appears in the set, then its weight is 1. The weight of a pattern in a set can be said to reflect the probability that the pattern belongs to that set.

If an attribute A is being examined at a node y_0 , with which set V_0 is associated, and if we are to do as described above, then patterns whose value of attribute A is missing in V_0 will not appear in any of the sets $V_1, V_2, \dots, V_n$, into which A splits the set V_0 . That will result in patterns with the missing values of attribute A not being associated with any of the descendants of y_0 . So, in practice, a fraction of each such pattern is added to the set associated with each child of y_0 , as described below.

1. For $1 \leq i \leq n$, let w_i be equal to the sum of the weights of the training patterns in set V_i .

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To evaluate $G_{\text{FOOTWEAR}}(V_0)$ at the middle child of the root of the decision tree given in Fig. 3.4, in which EATS has been examined at the root, the following procedure is used. The value of FOOTWEAR is missing for pattern T5. The middle child of the root is named y_0 . The classes are C_1 (professor) and C_2 (student). Therefore, $m = 2$. Since FOOTWEAR has two possible values (clogs and sandals), node y_0 has two children y_1 and y_2 . Therefore, $n = 2$.

$$\begin{aligned}
 V_0 &= \{T4, T5, T7, T8\} \text{ at node } y_0 \\
 Y(1,0) &= 2 \text{ (no. of known FOOTWEAR-value patterns in } V_0 \text{ of class } C_1) \\
 Y(2,0) &= 1 \text{ (no. of known FOOTWEAR-value patterns in } V_0 \text{ of class } C_2) \\
 Z(0) &= 4 \text{ (no. of patterns in } V_0) \\
 Z'(0) &= 3 \text{ (no. of known FOOTWEAR-value patterns in } V_0) \\
 V_1 &= \{T8\} \text{ at node } y_1, \text{ where FOOTWEAR} = \text{clogs} \\
 Y(1,1) &= 0 \text{ (no. of patterns in } V_1 \text{ of class } C_1) \\
 Y(2,1) &= 1 \text{ (no. of patterns in } V_1 \text{ of class } C_2) \\
 Z(1) &= 1 \text{ (no. of patterns in } V_1) \\
 V_2 &= \{T4, T7\} \text{ at node } y_2, \text{ where FOOTWEAR} = \text{sandals} \\
 Y(1,2) &= 2 \text{ (no. of patterns in } V_2 \text{ of class } C_1) \\
 Y(2,2) &= 0 \text{ (no. of patterns in } V_2 \text{ of class } C_2) \\
 Z(2) &= 2 \text{ (no. of patterns in } V_2) \\
 V_3 &= \{T5\} \text{ (patterns with missing value of FOOTWEAR)} \\
 Z(3) &= 1 \text{ (no. of patterns with missing value of FOOTWEAR)}
 \end{aligned}$$

$$I(V_0) = - \sum_{k=1}^m \frac{Y(k,0)}{Z'(0)} \log \frac{Y(k,0)}{Z'(0)} = 0.9183 \text{ [from Eqn (3.1)]}$$

$$I_{\text{FOOTWEAR}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z'(0)} \sum_{k=1}^m \frac{Y(k,j)}{Z(j)} \log \frac{Y(k,j)}{Z(j)} = 0 \text{ [from Eqn (3.2)]}$$

$$g_{\text{FOOTWEAR}}(V_0) = \frac{Z'(0)}{Z(0)} [I(V_0) - I_{\text{FOOTWEAR}}(V_0)] = 0.9183 \text{ [from Eqn (3.3)]}$$

$$S_{\text{FOOTWEAR}}(V_0) = - \sum_{j=1}^{n+1} \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 1.4466 \text{ [from Eqn (3.4)]}$$

$$G_{\text{FOOTWEAR}}(V_0) = \frac{g_{\text{FOOTWEAR}}(V_0)}{S_{\text{FOOTWEAR}}(V_0)} = 0.6351 \text{ [from Eqn (3.5)]}$$

FIGURE 3.5 Evaluating $G_{\text{FOOTWEAR}}(V_0)$, the ratio of information gain for FOOTWEAR, at the middle child of the root of the decision tree given in Fig. 3.4, in which EATS has been examined at the root. The value of FOOTWEAR is missing for pattern T5, and hence Eqns (3.1) to (3.5) are used above.

To evaluate $G_{\text{HABIT}}(V_0)$ at the middle child of the root of the decision tree given in Fig. 3.4, in which EATS has been examined at the root, the following procedure is used. The middle child of the root is named y_0 . The classes are C_1 (professor) and C_2 (student). Therefore, $m = 2$. Since HABIT has two possible values (gabby and quiet), node y_0 has two children y_1 and y_2 . Therefore, $n = 2$.

$$V_0 = \{T4, T5, T7, T8\} \text{ at node } y_0$$

$$Y(1, 0) = 2 \text{ (no. of patterns in } V_0 \text{ of class } C_1)$$

$$Y(2, 0) = 2 \text{ (no. of patterns in } V_0 \text{ of class } C_2)$$

$$Z(0) = 4 \text{ (no. of patterns in } V_0)$$

$$V_1 = \{T5, T7\} \text{ at node } y_1, \text{ where HABIT} = \text{gabby}$$

$$Y(1, 1) = 1 \text{ (no. of patterns in } V_1 \text{ of class } C_1)$$

$$Y(2, 1) = 1 \text{ (no. of patterns in } V_1 \text{ of class } C_2)$$

$$Z(1) = 2 \text{ (no. of patterns in } V_1)$$

$$V_2 = \{T4, T8\} \text{ at node } y_2, \text{ where HABIT} = \text{quiet}$$

$$Y(1, 2) = 1 \text{ (no. of patterns in } V_2 \text{ of class } C_1)$$

$$Y(2, 2) = 1 \text{ (no. of patterns in } V_2 \text{ of class } C_2)$$

$$Z(2) = 2 \text{ (number of patterns in } V_2)$$

$$I(V_0) = - \sum_{k=1}^m \frac{Y(k, 0)}{Z(0)} \log \frac{Y(k, 0)}{Z(0)} = 1.0 \text{ [from Eqn (2.1)]}$$

$$I_{\text{HABIT}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 1.0 \text{ [from Eqn (2.2)]}$$

$$S_{\text{HABIT}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 1.0 \text{ [from Eqn (2.4)]}$$

$$G_{\text{HABIT}}(V_0) = \frac{I(V_0) - I_{\text{HABIT}}(V_0)}{S_{\text{HABIT}}(V_0)} = 0 \text{ [from Eqn (2.5)]}$$

FIGURE 3.6 Evaluating $G_{\text{HABIT}}(V_0)$, the ratio of information gain of HABIT, at the middle child of the root of the decision tree given in Fig. 3.4, in which EATS has been examined at the root. No value of HABIT is missing, hence Eqns (2.1), (2.2), (2.4), and (2.5) are used above.

FIGURE

2. The sum of the weights is calculated as

$$\text{sum_of_weights} = \sum_{i=1}^n w_i$$

3. For every training pattern T in V_0 whose value of attribute A is missing, do steps 3.1 and 3.2.

- 3.1. Let f_0 be the fraction of pattern T present in V_0 .

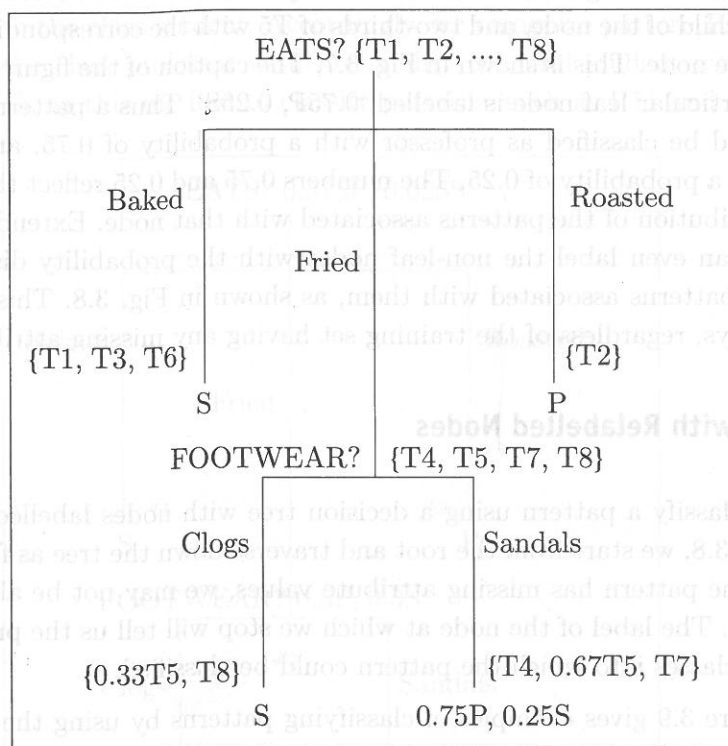


FIGURE 3.7 The decision tree showing the set of patterns associated with each node for the professor-student training set given in Fig. 3.1, in which the value of FOOTWEAR is missing for pattern T5. One-third (that is, 0.33) of T5 is associated with the left child of the node at which FOOTWEAR is examined. The left child is labelled S because the 1.33 patterns of T5 and T8 associated with it belong to the class student. Two-thirds (that is, 0.67) of T5 is associated with the right child of the node at which FOOTWEAR is examined. The right child is labelled '0.75P, 0.25S' because, of the 2.67 patterns associated with the child, two patterns (T4 and T7) belong to the class professor ($2/2.67 = 0.75$); and 0.67 pattern (namely T5) belongs to the class student ($0.67/2.67 = 0.25$). A pattern at this node will be classified as professor with a probability of 0.75, and as student with a probability of 0.25. You may want to compare this tree with the tree given in Fig. 2.7.

3.2. For $j = 1, 2, \dots, n$ do step 3.2.1.

3.2.1. Add to set V_j the f_j th fraction of pattern T , where

$$f_j = \frac{w_j}{\text{sum_of_weights}} f_0$$

4. Return from the procedure.

If we are to apply this procedure to the node at which FOOTWEAR is examined in Fig. 3.4, then one-third of pattern T5 will be associated with the left child of the node, and two-thirds of T5 with the corresponding right child of the node. This is shown in Fig. 3.7. The caption of the figure explains how a particular leaf node is labelled '0.75P, 0.25S'. Thus a pattern at this node would be classified as professor with a probability of 0.75, and as student with a probability of 0.25. The numbers 0.75 and 0.25 reflect the probability distribution of the patterns associated with that node. Extending this idea, we can even label the non-leaf nodes with the probability distributions of the patterns associated with them, as shown in Fig. 3.8. This can be done always, regardless of the training set having any missing attribute values.

3.2 Classifying with Relabelled Nodes

To classify a pattern using a decision tree with nodes labelled as shown in Fig. 3.8, we start from the root and traverse down the tree as far as possible (if the pattern has missing attribute values, we may not be able to reach a leaf). The label of the node at which we stop will tell us the probabilities of the classes into which the pattern could be classified.

Figure 3.9 gives examples of classifying patterns by using the decision tree shown in Fig. 3.8. The nodes have been named $x, x_1, x_2, \dots$ in the decision tree to make the figure easier to understand. The figure contains all the training patterns given in Fig. 3.1 and a few recall patterns. The figure shows that the classifications of few of the training patterns by the decision tree given in Fig. 3.8 are not the same as that given in Fig. 3.1.

Three patterns—R11, R12 and R13—given in Fig. 3.9 need comments, because for each of them we are unable to traverse down the tree: we have to stop at the root. In R11, the only attribute value known is FOOTWEAR equal to clogs, so from the label of the root x , we say that the pattern belongs to the class professor with a probability of 0.375, and it belongs to

the class student with a probability of 0.625. In a situation, where we have to stop at the root itself, an alternative method can be used to classify the pattern. In the training set given in Fig. 3.1, two patterns, T1 and T8, have FOOTWEAR equals to clogs, and both belong to the class student. So we recognize pattern R11 as belonging to the class student.

Similarly, for pattern R12 (HABIT = gabby), we notice in the training set given in Fig. 3.1, that out of the five patterns with HABIT equal to gabby, two (T2 and T7) belong to the class professor, and three (T1, T3, T5) belong to the class student. Accordingly, we recognize pattern R12 as belonging to the class professor with probability 0.4, and student with probability 0.6. Using this method to classify patterns, such as R11 and R12, requires that

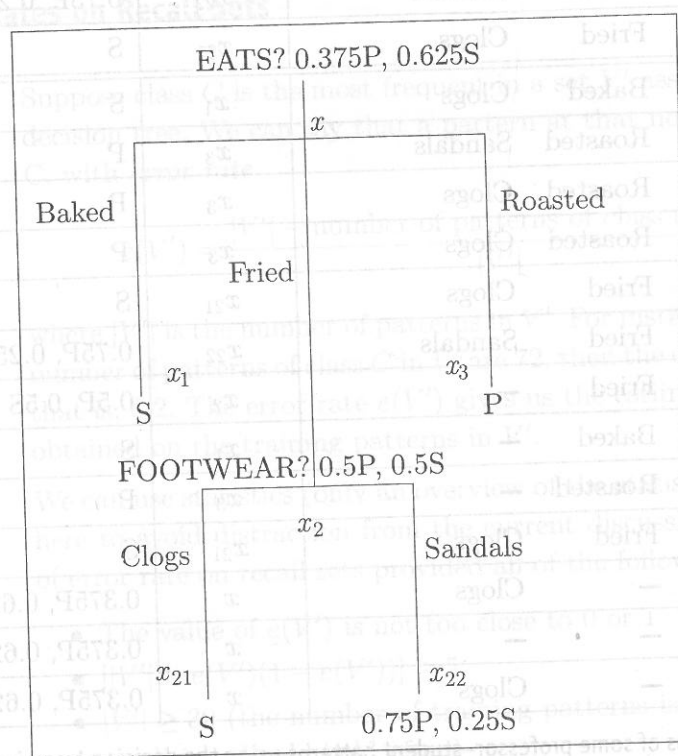


FIGURE 3.8 The decision tree obtained from Fig. 3.7 with each node labelled with the probability distribution of the classes of the patterns associated with that node. The tree is for the professor-student training set given in Fig. 3.1, in which the value of FOOTWEAR is missing for pattern T5. Once we have labelled the nodes as above, we need not show the patterns associated with each node. The nodes have been named $x, x_1, x_2, \dots$ to make Fig. 3.9 easier to understand.

NAME of pattern	Attributes			Node of Fig. 3.8 reached	Classification
	HABIT	EATS	FOOTWEAR		
T1	Gabby	Baked	Clogs	x_1	S
T2	Gabby	Roasted	Sandals	x_3	P
T3	Gabby	Baked	Sandals	x_1	S
T4*	Quiet	Fried	Sandals	x_{22}	0.75P, 0.25S
T5*	Gabby	Fried	—	x_2	0.5P, 0.5S
T6	Quiet	Baked	Sandals	x_1	S
T7*	Gabby	Fried	Sandals	x_{22}	0.75P, 0.25S
T8	Quiet	Fried	Clogs	x_{21}	S
R1	Quiet	Baked	Clogs	x_1	S
R2	Quiet	Roasted	Sandals	x_3	P
R3	Gabby	Roasted	Clogs	x_3	P
R4	Quiet	Roasted	Clogs	x_3	P
R5	Gabby	Fried	Clogs	x_{21}	S
R6	—	Fried	Sandals	x_{22}	0.75P, 0.25S
R7	—	Fried	—	x_2	0.5P, 0.5S
R8	—	Baked	—	x_1	S
R9	—	Roasted	—	x_3	P
R10	—	Fried	Clogs	x_{21}	S
R11	—	—	Clogs	x	0.375P, 0.625S
R12	Gabby	—	—	x	0.375P, 0.625S
R13	Gabby	—	Clogs	x	0.375P, 0.625S

FIGURE 3.9

Classifications of some professor-student patterns using the decision tree given in Fig. 3.8. A missing value of an attribute is shown by a dash. Patterns T1 to T8 are the training patterns taken from Fig. 3.1. An asterisk besides the name of a training pattern indicates that its classification here is not identical to that in its training set given in Fig. 3.1. Patterns R1 to R13 are recall patterns, of which R1 to R4 are the same as those in Fig. 1.2. The classifications for patterns R11 to R13 have been obtained from the label at the root of the decision tree. For such patterns, an alternative method of classification is discussed in Section 3.2.

the class distributions for each attribute value and combinations of attribute values in the training set be available at the time of classification.

We would need the class distribution for attribute value combinations for pattern R13 (HABIT = gabby and FOOTWEAR = clogs). According to the training set given in Fig. 3.1, only one pattern, namely T1, has HABIT equal to gabby and FOOTWEAR equal to clogs; and it is classified as student, so we would recognize R13 also as belonging to the class student. The method (the one used in Fig. 3.9 or the one discussed here) you will use for patterns such as R11 to R13 is your design choice. As is clear, the two methods may not give the same results.

3.3 Error Rates on Recall Sets

Suppose class C is the most frequent in a set V' associated with a node in a decision tree. We can say that a pattern at that node will be classified into C , with *error rate*

$$e(V') = \frac{|V'| - \text{number of patterns of class } C \text{ in } V'}{|V'|}$$

where $|V'|$ is the number of patterns in V' . For instance, if $|V'| = 90$, and the number of patterns of class C in V' are 72, then the error rate is $(90 - 72)/90$, that is, 0.2. The error rate $e(V')$ gives us the estimated probability of error obtained on the training patterns in V' .

We can use statistics (only an overview of the statistics involved is described here to avoid distraction from the current discussion) to predict the range of error rate on recall sets provided all of the following conditions are true:

- The value of $e(V')$ is not too close to 0 or 1
- $[|V'| \times e(V')(1 - e(V'))] \geq 5$
- $|V'| \geq 30$ (the number of training patterns is usually large, hence this condition should hold often)
- The patterns in V' are independent of one another
- The recall sets will be drawn from the same population of patterns as V'

When the range of error rate for $0 \leq N \leq 100$ per cent of the recall sets is predicted, it is said to be prediction with N per cent confidence. Under

Fig. 3.8.
training
indicates
Fig. 3.1.
Fig. 1.2.
the root
scussed

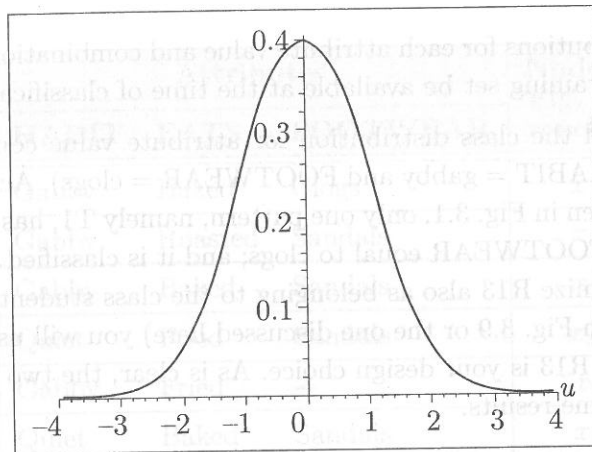


FIGURE 3.10 Normal distribution curve, obtained by plotting $\frac{1}{\sqrt{2\pi}}e^{-u^2/2}$.

the above conditions, the range of error rate can be approximated to have a Normal (or Gaussian) distribution (Fig. 3.10), which is plotted by the function

$$\frac{1}{\sqrt{2\pi}}e^{-u^2/2}$$

The area under the curve from $u = -\infty$ to $u = \infty$ is 1, denoting, for this purpose, 100 per cent confidence. For a given value of N per cent confidence, we want to find the value of z such that the area under the curve from $u = -z$ to $u = z$ is $N/100$. Since the curve is symmetric, this is equivalent to finding the value of z such that the area under the curve from $u = 0$ to $u = z$ is $N/200$. This can be done by solving for z in the equation

$$\frac{1}{\sqrt{2\pi}} \int_0^z e^{-u^2/2} du = \frac{N}{200}$$

We can write a computer program that, given a value of N , will find the value of z from the above equation. Alternatively, we can find the value of z corresponding to N from Fig. 3.11. Once we have selected a value of N and found the corresponding value of z , we can predict—according to a statistical result—that for N per cent of the recall sets, the error rate on a

$N(\%)$	z
0	0.00
1	0.00
2	0.00
3	0.00
4	0.00
5	0.00
6	0.00
7	0.00
8	0.1
9	0.1
10	0.1
11	0.1
12	0.1
13	0.1
14	0.1
15	0.1
16	0.1

FIGURE 3

$N(\%)$	z	$N(\%)$	z	$N(\%)$	z	$N(\%)$	z	$N(\%)$	z	$N(\%)$	z
0	0.00	17	0.21	34	0.44	51	0.69	68	1.00	85	1.44
1	0.01	18	0.23	35	0.45	52	0.71	69	1.02	86	1.48
2	0.03	19	0.24	36	0.47	53	0.73	70	1.04	87	1.51
3	0.04	20	0.25	37	0.49	54	0.74	71	1.06	88	1.56
4	0.05	21	0.27	38	0.50	55	0.76	72	1.08	89	1.60
5	0.06	22	0.28	39	0.51	56	0.77	73	1.10	90	1.64
6	0.08	23	0.29	40	0.52	57	0.79	74	1.13	91	1.70
7	0.09	24	0.31	41	0.53	58	0.81	75	1.15	92	1.75
8	0.10	25	0.32	42	0.55	59	0.82	76	1.18	93	1.81
9	0.11	26	0.33	43	0.57	60	0.84	77	1.20	94	1.88
10	0.13	27	0.34	44	0.58	61	0.86	78	1.23	95	1.96
11	0.14	28	0.36	45	0.60	62	0.88	79	1.25	96	2.05
12	0.15	29	0.37	46	0.61	63	0.90	80	1.28	97	2.17
13	0.17	30	0.39	47	0.63	64	0.92	81	1.31	98	2.33
14	0.18	31	0.40	48	0.64	65	0.93	82	1.34	99	2.58
15	0.19	32	0.41	49	0.66	66	0.96	83	1.37	99.99	3.90
16	0.20	33	0.43	50	0.67	67	0.97	84	1.41	100	∞

FIGURE 3.11 Values of z corresponding to confidence N varying from 0 to 100 per cent. For a given N per cent, the area under the normal distribution curve (Fig. 3.10) between $u = 0$ and $u = z$ is $N/200$. Thus, for a given N , the value of z can be obtained from the equation

$$\frac{1}{\sqrt{2\pi}} \int_0^z e^{-u^2/2} du = \frac{N}{200}.$$

given recall set will lie in the interval

$$e(V') \pm z \sqrt{\frac{e(V')[1 - e(V')]}{|V'|}}$$

Suppose we want to predict the error rate with 95 per cent confidence for the above example, where $|V'|$ is 90, and $e(V')$ is 0.2. We see from Fig. 3.11

that, for $N = 95$, the value of z is 1.96. Substituting these values in the above equation, we can predict that, for 95 per cent of the recall sets, the error rate will lie in the interval

$$\begin{aligned} & 0.2 \pm 1.96\sqrt{0.2(1-0.2)/90} \\ & = 0.2 \pm 0.08 \\ & = [0.12, 0.28] \end{aligned}$$

The square brackets, as is conventional in mathematics, represent a closed interval. From the above, we can make the following predictions.

- For 95 per cent of the recall sets, the error rate will be at least 0.12 and at the most 0.28.
- For 2.5 per cent of the recall sets (that is, with 2.5 per cent confidence), the error rate will be less than 0.12. Thus, for 97.5 per cent of the recall sets (that is, with 97.5 per cent confidence), the error rate will be 0.12 or more.
- For 2.5 per cent of the recall sets (that is, with 2.5 per cent confidence), the error rate will be more than 0.28. Thus, for 97.5 per cent of the recall sets (with 97.5 per cent confidence), the error rate will be 0.28 or less.

The upper bound of the interval $[0.12, 0.28]$ can be considered to be the upper bound of the error rate on recall sets. Thus, the upper bound of the error rate for recall sets can be considered to be 0.28.

3.4 Pruning Decision Trees

In a decision tree, consider a node x_i whose children x_{i1} to x_{in} are leaf nodes. If the upper bound of the predicted error rate at x_i is lower than the upper bounds of the predicted error rates at each of its children x_{i1} to x_{in} , then—as a design choice—we can delete the children x_{i1} to x_{in} . This is known as *pruning* the tree. Node x_i then becomes a leaf node. We can repeat this process, thus iteratively pruning the decision tree. A pruned tree, being shallower and thus examining fewer attributes, is expected to classify patterns faster than an unpruned tree. Moreover, it has often been found in practice that pruned decision trees perform better on recall sets than unpruned trees: deep, unpruned decision trees tend to memorize the training set. They are said to be *overfitted* to the training set, that is, they

become too specific to the training set. Pruned trees generalize better and so perform better on recall sets.

A node in a decision tree, be it unpruned or pruned, can be labelled with the class into which a given pattern will be classified at that node, together with the error rate $e(V')$ and the upper bound of the predicted error rate (if such prediction was made because the conditions listed in Section 3.3 were true at that node). That way, when the tree is used to classify some given pattern, the user of the decision tree can know not only the pattern's classification, but also the error rate $e(V')$ and the upper bound of the predicted error rate. All three—the classification, the error rate $e(V')$, and the upper bound of the error rate—are taken from the label of the node at which the pattern is classified. A user may choose to have acceptable error rates, depending on the domain of patterns—the acceptable error rates would understandably be lower for a medical domain than for recognizing, say, professors and students. For a given pattern, if any of the error rates provided by the decision tree is higher than what the user considers acceptable, then the user may consider the pattern rejected, that is, the decision tree has failed to recognize the pattern. It is the user who then determines when a pattern can be considered as rejected by the decision tree.

3.5 Windowing of Training Sets

For building decision trees, which perform well on recall patterns, large training sets are required. Large training sets, however, slow down the process of building the decision tree. One approach, which sometimes works in practice, is to select a proper subset W of V , the large training set. W is called the *window*. The class distribution of the patterns in W could be the same as in V , or the classes in W may be equiprobable.

Build a decision tree from W . Classify the patterns of $(V - W)$ with the decision tree built. If the error exceeds an acceptable limit, add the misclassified patterns to W . Rebuild the decision tree on the enlarged W . Repeat the above process till the error rate falls below the acceptable limit. This, of course, is not guaranteed to happen, and in the worst case, the full V will be needed to build the decision tree. One may choose to try building the decision tree by such windowing of the training set.

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Summary

At times, patterns in a training set may have some of the attribute values missing. To build a decision tree from such a training set, the equation to maximize the ratio of information gain is modified. Such a tree can have fractions of patterns associated with a node. Each node can be labelled with the class probability distribution of the patterns associated with the node. From the distributions, we can get the error rate on the training patterns, which can be used to predict the error rates on recall sets. For very large training sets, decision trees can be built by windowing the training sets.

Exercises

1. Modify the program you wrote either in the first exercise or second exercise of Chapter 2 to build a decision tree with some attribute values missing. With the observed error on the training set, the program should be able to predict the range of error rates on recall sets at a given confidence level. For a given confidence level N , the program should read the value of z from Fig. 3.11 stored in computer memory, or it could calculate the value of z from the equation

$$\frac{1}{\sqrt{2\pi}} \int_0^z e^{-u^2/2} du = \frac{N}{200}$$

To find the value of z from the above equation, the knowledge of procedures for numerical integration and for finding roots of equations is required.

2. With the program you wrote in the first exercise, build a decision tree when the value of HABIT is missing from pattern T3 of the training set given in Fig. 1.1. Use the tree to classify the patterns listed in Fig. 3.9. Predict the range of error rates on recall sets at confidence levels of ninety, ninety-five, and ninety-nine per cent. Although the conditions required to predict error rates (listed in Section 3.3) will not be true, you may assume they are true for this exercise.
3. Repeat the second exercise when the value of HABIT is missing from pattern T3, and the value of EATS is missing from pattern T7 of the training set given in Fig. 1.1.

4. In a constituency of 200,000 voters, suppose you interview 1000 arbitrarily selected people and find out that, in the upcoming election, 53 per cent plan to vote for the Revolutionary party and the remaining for the Rotationary party. Based on Section 3.3, can you predict with 99 per cent confidence that, in the election, the percentage of votes for the Revolutionary party will lie in the interval $[49, 57]$, and for the Rotationary party in the interval $[43, 51]$? If you had, instead, interviewed 10,000 people, and found that the same percentages planned to vote for each of the two parties, then with 99 per cent confidence find the corresponding two voting intervals. Have the intervals narrowed or widened? What would be the intervals if you carried out the last step at 95 per cent confidence, instead of 99 per cent?

Obtaining Prules
by Evolution